Spherical Geometry

Narnia Poddar

1 Introduction

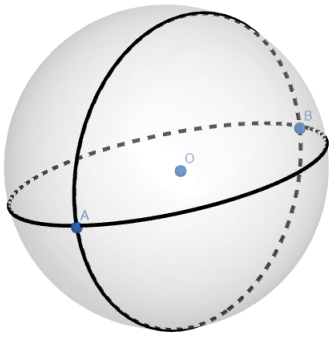
We're all familiar with Euclidean geometry – geometry on a plane – but not so much with geometry on a sphere like the Earth. Though spherical geometry still follows the first four axioms that Euclidean geometry is governed by, it doesn't abide by the fifth one: the existence of parallel lines (which Euclid initially thought could be proved by the other four, but later discovered couldn't)! This may not seem like a great change, but it leads a geometry on the surface of a sphere to have very different properties than those we are used to seeing on a plane.

2 Lines and Distances

On a sphere, we define a line to be a great circle. A great circle is a circle that lies on the surface of the sphere and whose center is at the center of the sphere (and thus has the same radius as the sphere). We chose great circles to be our lines rather than any other circle on the surface because the distances between two points on the surface of the sphere are minimized by traveling on a great circle.

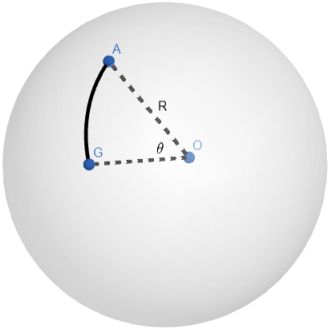
From our definition of a line, we can also find that there are no parallel lines on a sphere, as every pair of great circles intersects at two antipodal points.

Figure 1: A pair of great circles intersecting at two antipodal points (diagram made with Geogebra)



Finally, we define the distance between two points (also known as a geodesic) to be shortest arc of a great circle connecting the points. Since all great circles have same radius as the sphere and are centered at the sphere's center, the angle subtended by the geodesic uniquely determines the length of the geodesic. The length of a geodesic with central angle θ can therefore be computed as $\frac{\theta}{2\pi} \cdot 2\pi R = R\theta$.

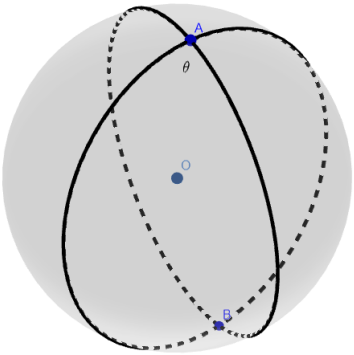
Figure 2: A geodesic



3 Bigons

We've already observed that any two different lines intersect in exactly two antipodal points of the sphere, and they split the sphere into several closed regions. Each of these regions only have two edges, meaning we can have two sided polygons – bigons – in spherical geometry.

Figure 3: A bigon with antipodal points A and B



To find the area of a bigon, we want to know what fraction of the surface area of the sphere it takes up. Since the two points of intersection of the bigons are antipodal, we can rotate the sphere so that the two endpoints of the bigon are the northmost and southmost points (the poles) of the sphere. The planes of the great circle that define the bigon intersect along the line joining the north and south pole. Suppose the angle between them is θ. Then the bigon makes up $\frac{\theta}{2\pi}$ of the surface area of the sphere, meaning it has area $\frac{\theta}{2\pi} \cdot 4\pi R^2 = 2R^2\theta$.

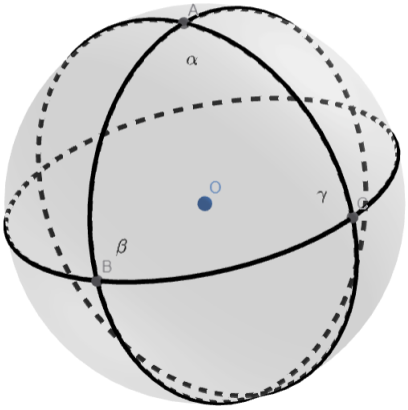
A slightly easier way to find the area is to measure the angle where the two great circles intersect on the surface of the sphere. We know this angle is the same as θ because the great circles intersect at the same angle as the planes they determine intersect at. From here, we use a similar logic to determine that the area is $\frac{\theta}{2\pi} \cdot 4\pi R^2 = 2R^2\theta$.

4 Triangles

4.1 Computing areas

Suppose we have a triangle ABC on the sphere, with $\angle BAC = \alpha$, $\angle ABC = \beta$, and $\angle BCA = \gamma$, as shown in figure 4 below.

Figure 4: A triangle



Let the area of the triangle be A. If we take the union of the three bigons determined by each pair of lines, we get half the area of the sphere (since for every point in a bigon, its antipodal point is not included in a bigon). This means the union of the areas is half the area of the sphere, or $\frac{1}{2} \cdot 4\pi r^2 = 2\pi r^2$.

The union of the areas of the bigons can also be written as the sum of the areas of each bigon, minus two times the area of the triangle as it is overcounted twice (it is included once in each bigon). Equating the two expressions for the union of the areas of the bigons:

$$\begin{aligned} 2\alpha R^2 + 2\beta R^2 + 2\gamma R^2 - 2A &= 2\pi R^2 \\ 2A &= 2\alpha R^2 + 2\beta R^2 + 2\gamma R^2 - 2\pi R^2 \\ A &= R^2(\alpha + \beta + \gamma - \pi) \end{aligned}$$

4.2 More about angles

We know that area is always positive, so applying this to our area formula for triangles:

$$\begin{aligned} R^2(\alpha + \beta + \gamma - \pi) &> 0 \\ \alpha + \beta + \gamma - \pi &> 0 \\ \alpha + \beta + \gamma &> \pi \end{aligned}$$

Therefore, the sum of the angles in a triangle is always greater than π.

5 A new congruence theorem

Like in Euclidean geometry, SSS, SAS, and ASA congruence hold true on a sphere. An interesting result of this is a new spherical congruence theorem: AAA (which had only guaranteed similarity in Euclidean geometry)! For simplicity, we assume we are working with a unit sphere ($R = 1$) in this section.

5.1 Poles

First, we define a pole of a line l to be one of the two points where the line going through the center and perpendicular to the plane of the great circle of l intersects the sphere. We observe the following properties of poles:

1. The length PA of the geodesic connecting pole P of line to any point A on line l is $\frac{1}{4} \cdot 2\pi = \frac{\pi}{2}$ since the line connecting the pole to the center is perpendicular to the plane of the line (and because we are assuming $R = 1$ for this section).

2. Given a line segment AB and a point P such that $PA = \frac{\pi}{2}$ and $PB = \frac{\pi}{2}$, then P is a pole of AB because $PA = PB = \frac{\pi}{2}$ implies that PO is perpendicular to the great circle containing AB.

3. The length of the geodesic between two points A and B is $\angle APB$, because $\angle APB \cong \angle AOB$.

5.2 Polar triangles

We define the polar triangle of $\triangle ABC$ to be $\triangle A'B'C'$ such that A' is a pole of BC on the same side of BC as A, B' is a pole of AC on the same side of AC as B, and C' is a pole of AB on the same side of AB as C. Using the three properties above, we can prove the following theorems:

1. Given the polar triangle of $\triangle ABC$, $\triangle A'B'C'$, $\angle A + B'C' = \pi$.

   To prove this, we extend AB and AC to meet $B'C'$ at points X and Y. We then have that $B'Y = C'X = \frac{\pi}{2}$ by property 1 of section 5.1. $B'Y + C'X = B'C' + XY$, so $B'C' + XY = \frac{\pi}{2} + \frac{\pi}{2} = \pi$. By property 3 of section 5.1, we know that $\angle A = XY$. Plugging this into the equation, we get $\angle A + B'C' = \pi$, as desired.

2. If $\triangle A'B'C'$ is the polar triangle to $\triangle ABC$, then $\triangle ABC$ is also the polar triangle of $\triangle A'B'C'$.

   We know B' is a pole of AC, so by property 1 of section 5.1, $AB' = \frac{\pi}{2}$. By a similar logic, $AC' = \frac{\pi}{2}$, which implies that A is a pole of $B'C'$. Similarly, we can show that B is a pole of $A'C'$ and C is a pole of $A'B'$.

5.3 AAA congruence

Let $\triangle ABC$ and $\triangle DEF$ have $\angle A \cong \angle D$, $\angle B \cong \angle E$, and $\angle C \cong \angle F$. Consider their polar triangles, $\triangle A'B'C'$ and $\triangle D'E'F'$. By theorem 1 of section 5.2:

$$B'C' + \angle A = \pi$$
$$E'F' + \angle D = \pi$$

Since $\angle A \cong \angle D$ by assumption, we know $B'C' = E'F'$ and can similarly find that $A'B' = D'E'$ and $A'C' = D'F'$. Therefore, by SSS congruence, we know that $\triangle A'B'C' \cong \triangle D'E'F'$.

We know that by theorem 2 of section 5.2, $\triangle ABC$ is the polar triangle of $\triangle A'B'C'$ and $\triangle DEF$ is the polar triangle of $\triangle D'E'F'$. Applying theorem 1 of section 5.2 again:

$$BC + \angle A' = \pi$$
$$EF + \angle D' = \pi$$

Since we've shown that $\triangle A'B'C' \cong \triangle D'E'F'$, we know that $\angle A' \cong \angle D'$, so $BC = EF$. Similarly, we find that $AB = DE$ and $AC = DF$, which tells us that $\triangle ABC \cong \triangle DEF$ by SSS.

References

Arapura, Donu. "MA 460 Supplement: spherical geometry." Purdue Math, https://www.math.purdue.edu/~arapura/460/spherical.pdf.

Bolin, Michael."Exploration of Spherical Geometry." 9 September 2003, https://math.iit.edu/~mccomic/420/notes/Bolin_spherical.pdf.

Whittlesey, Marshall A. "Axiomatic spherical geometry." Spherical Geometry and its Applications, CRC Press, 2020, https://sidoli.w.waseda.jp/Whittlesey_2020_Spherical_Geometry_Chap_3.pdf.